1 **(a)** A beaker in air contains a liquid. The base area of the beaker is A, as shown in Fig. 1.1. The liquid has density ρ and fills the beaker to a height h.

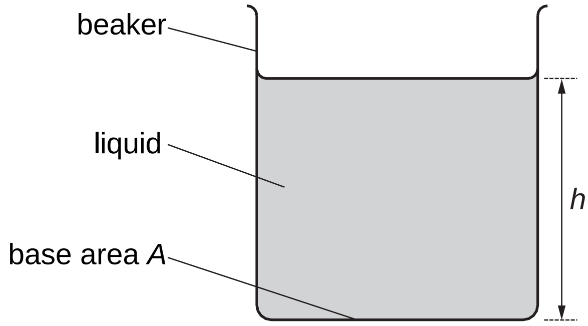

Fig. 1.1

(i) Show that the pressure P due to the liquid at the base of the beaker is given by

$$P = \rho g h$$

where g is the acceleration of free fall.

$P = \dfrac{F}{A}$ and $\rho = \dfrac{m}{Ah}$ both relations used

$$P = \frac{m \times g}{A} = \frac{\rho A h \times g}{A} = \rho g h$$

appropriate algebra leading to ans [B1] [1]

(ii) Explain why the equation in **(i)** does not give the total pressure at the base of the beaker.

Total pressure at the base includes atmospheric / air pressure above the liquid [B1]

[1]

(iii) Fig. 1.2 shows the variation of the total pressure inside the liquid with depth x below the surface.

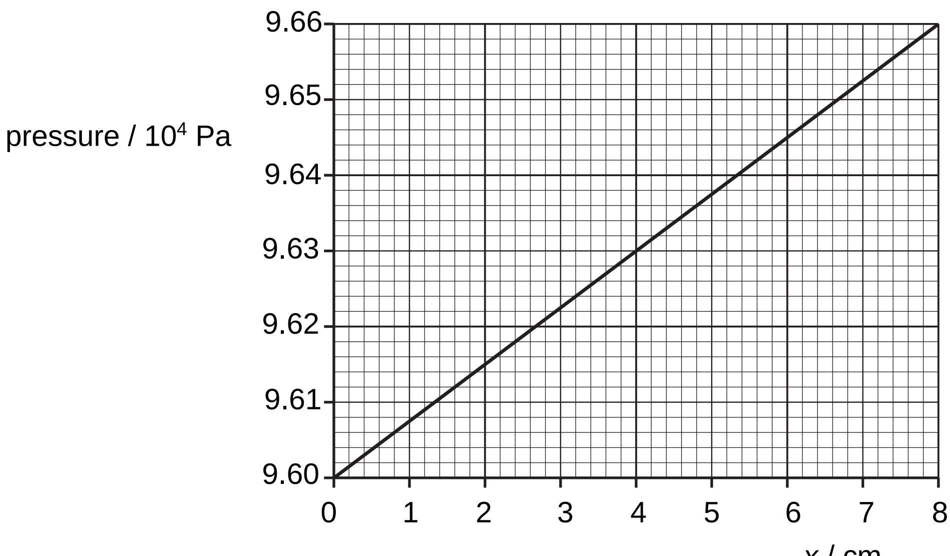

Fig. 1.2

Use Fig. 1.2 to determine the density of the liquid.

$$p_{total} = \rho g h + p_{atm}$$

$$9.66 \times 10^4 = \rho(9.81)(0.080) + 9.60 \times 10^4$$

$$\rho = 765 \text{ kg m}^{-3} \quad \text{[A1] (accept 760 to 770 kg m}^{-3}\text{)}$$

$$\Delta p = \rho g \Delta h$$

$$(9.66 - 9.60) \times 10^4 = \rho(9.81)(8.0 \times 10^{-2})$$

$$\rho = 765 \text{ kg m}^{-3} \quad \text{[A1] (accept 760 to 770 kg m}^{-3}\text{)}$$

density = ...760 to 770.......... kg m^{-3} [1]

(b) A spherical buoy of density 220 kg m^{-3} floats in equilibrium on the surface of sea water of density 1050 kg m^{-3}, as shown in Fig. 1.3.

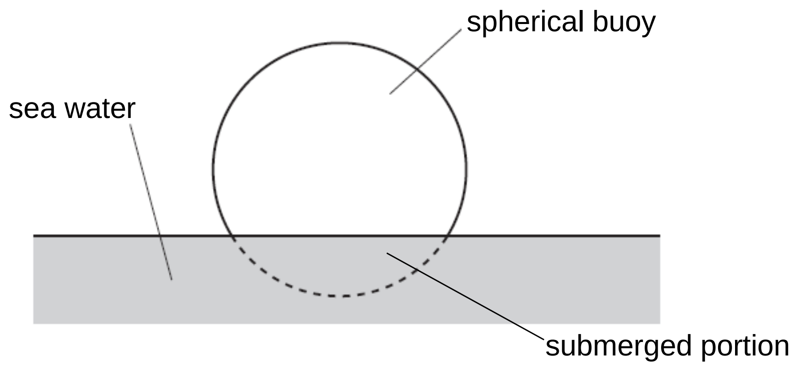

Fig. 1.3 (not to scale)

Determine the percentage of the volume of the buoy that is submerged in water.

At equilibrium during floating, upthrust = weight

Let submerged volume be V_{sub} and volume of sphere be V.

$$(\rho_W)(V_{sub})g = (\rho_S)(V)g$$

$$1050 V_{sub} = 220 V \quad \text{[C1]}$$

$$\frac{V_{sub}}{V} = 0.21$$

$$\therefore \text{percentage submerged} = 21\% \quad \text{[A1]}$$

percentage = ...21.......... % [2]

[Total: 5]

2 **(a)** A body travelling at a constant speed in a circular path experience centripetal acceleration.